H. Şeyda Gündüz Jordan

Engineering Leader • Software Architecture & AI Systems
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Profile

Engineering leader with 15+ years of experience building production software, designing system architectures, and leading engineering teams. Combines deep expertise in software architecture, observability, and engineering quality, with hands on AI development and research in deep learning, computer vision and retrieval-augmented generation. Currently completing an MSc in Informatics, majoring in Artificial Intelligence, at the University of Zurich.

Selected AI and Software Projects

MRI-to-Synthetic-CT Generation with Diffusion Models *UZH & University Hospital Zurich, 2025*

- Designed and implemented an end-to-end conditional diffusion pipeline for synthetic CT generation from paired MRI–CT data, covering data analysis, preprocessing, conditional diffusion model training and DDIM sampling for accelerated inference.
- Developed the codebase and co-authored the research report.

Trajectory Aware Diverse Memory Selection *Independent research, 2026*

- Investigated geometric diversity as a complement to relevance in conversational memory selection under token constraints.
- Built a benchmarking framework comparing multiscale metric space magnitude with relevance-only retrieval and MMR, using datasets with curated conversational memories across embedding models and token budgets. codebase

RadConductor – Medical-imaging workflow orchestration *Independent project, 2026–present*

- Developing a local first medical imaging pipeline with support for external analysis models, towards an extensible framework where multiple imaging models, analytical tools, and LLM agents work together. codebase

Professional Experience

Toptal Remote
Quality Engineering Manager (Contract) *2019–2024*

Mode · *2024*

- Aligned product, design, development and QA teams through BDD on the expected behaviour of an end-to-end encrypted backup communication channel designed to maintain company communications during cyber incidents.
- Built the test automation framework and trained internal QA teams to extend it.

InteliCare · *2021–2022*

- Took ownership of an undocumented production platform built on asynchronous Azure Functions, reverse-engineering its behaviour and requirements.
- Implemented distributed tracing across 10+ services, built the automated test suite, and trained the development team in quality and product practices before handover.

Lykke · *2019–2020*

- Served as quality lead across a 50+ person programme building a blockchain-based property trading and loyalty platform for Emaar Group.
- Established engineering workflows, observability, release pipelines and quality controls to support consistent delivery.
- Built the initial test automation framework and grew the QA team to six engineers, establishing automated end-to-end validation across the platform's microservices, enabling developers to make changes confidently.

Cruit World OÜ Remote
Founder, part-time *2021–2023*

- Took an AI-assisted hiring platform from concept to working MVP, defining the product roadmap and building CV parsing, semantic job matching, candidate ranking and API integrations.
- Conducted product discovery with job seekers, recruiters, and hiring managers to validate user needs and refine the product direction.

Aurea Software Remote
Test Automation Chief Software Architect *2018–2019*

- Designed the test automation architecture for a customer support platform rewrite, defining framework conventions, CI/CD quality gates and performance benchmarks.

Comodo Software Solutions Ankara, Turkey
Test Automation Architect *2017–2018*

- Architected the organisation's test automation framework and guided its adoption across multiple QA teams.
- Trained 20+ quality engineers in software development and test automation, supporting their transition into active test automation engineering roles.

T2 Ankara, Turkey
Senior QA and Software Engineer *2015–2017*

- Designed and delivered BDD workshops for one of Turkey's largest banks, and advised transformation leadership on adoption.
- Integrated new sensors into a central station system as a software engineer on a national defence project.
- Worked with Turk Telekom stakeholders to clarify requirements and resolve technical and delivery trade-offs.

Innova IT Solutions Ankara, Turkey
Test Automation Consultant *2014–2015*

- Automated the full end-to-end testing process: read requirements from BPMN XML files, generated and ran API tests with automatic reporting, and verified results against the UI's process history.
- Cut the effort of each requirements change from 5–6 days of manual API test updates by 11 people to an automated run of about 90 minutes.

Various Organisations International
Test Engineer *2009–2014*

- Tested enterprise and public-sector software, including the website of the NYC Department of Education and Microsoft product localisation through Lionbridge.

Education

University of Zurich Zurich, Switzerland
MSc Informatics – AI Major; Data Science Minor *2024–February 2027 (expected)*
Coursework includes deep learning, advanced machine learning, RAG, vision algorithms, reinforcement learning, probability theory, real analysis, human-centred AI.

Bilkent University Ankara, Turkey
Non-degree graduate coursework – Machine Learning *2015*

Middle East Technical University Ankara, Turkey
BSc in Statistics *2006–2010*

Technical Skills

AI and machine learning: Python, PyTorch, deep learning, diffusion models, computer vision, RAG
Software and systems engineering: Software architecture, microservices, event driven and serverless architectures, Java, TypeScript, BDD
Production and quality engineering: CI/CD, observability, distributed tracing, monitoring, test automation architecture, performance testing